BayesINE1. Introducción. Métodos Bayesianos y Estadística Oficial

Intro a Estadística Bayesiana

DataLab

Abril 2026

ICMAT-CSIC

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Course Objectives

Course objectives

- Introduce main concepts in Bayesian inference
- Illustrate them with several examples from official stats
- Illustrate key concepts with the aid of R (and several related packages, including Stan)
- Enable self-transition towards probabilistic machine learning

Course Contents

Course contents

1. Intro. Bayesian methods and official stats
2. Estimation of proportions. Beta-binomial model. Formalization
3. Estimation of means. Normal-normal model. Formalization and recap
4. Linear regression. Hierarchical models. Empirical Bayes methods.
5. Time series models. Dynamic linear models
6. Generalised linear models. Logistic regression.
7. Computational methods in Bayesian stats. Introduction
8. Decision Analysis
9. Some applications in official stats

Bayesian Stats and Official Stats

Bayesian methods and official stats

- Official Stats pervaded by frequentist methods
  - Comparatively little presence in INE's official requirements
  - Extremely little presence in Foundations and Advances of Machine Learning in Official Statistics

Bayesian methods and official stats

BayOff report

- Computations
- Subjectivism
- Imprecision and sensitivity to the prior
- The tradition in Official Stats

Bayesian methods and official stats

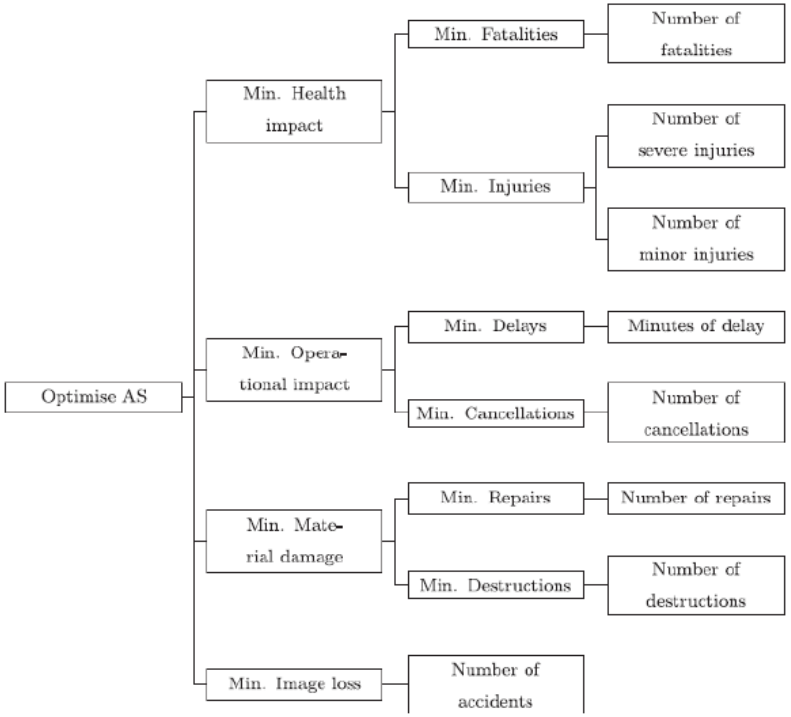
The times they are'a-changin'.....

- FDA Issues Guidance on Modernizing Statistical Methods for Clinical Trials
- In the prelim versions of the EU AI Act
- Calibrated Bayes

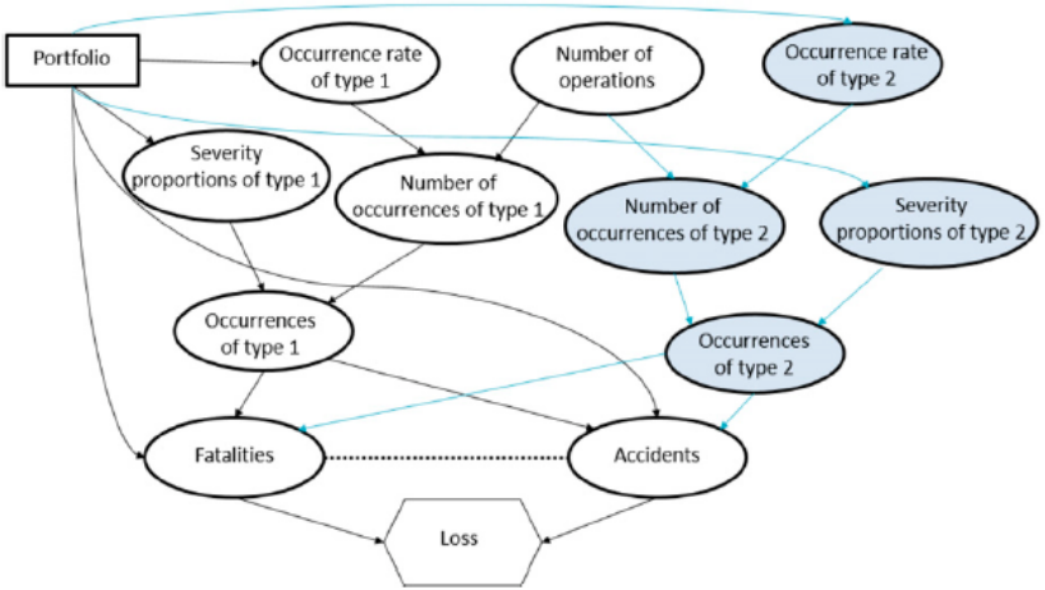
Bayesian methods and official stats. RIMAS @AESA

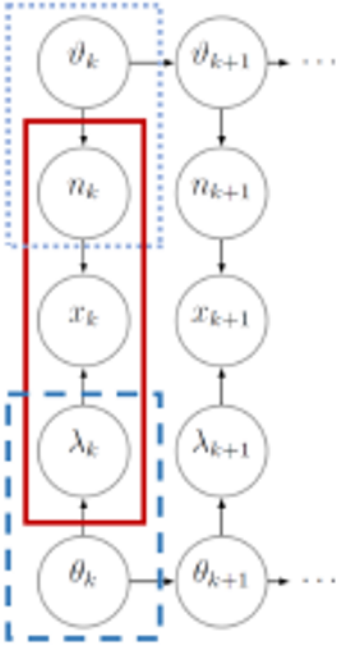
Bayesian methods and official stats. RIMAS @AESA

- 2.5 M operations
- 50 airports
- 250 airfields
- 44 Airlines
- 7000 aircraft
- ...
- Approx. 30000 incidents, 40 accidents
- 88 types of occurrences
- 8 consequences of interest



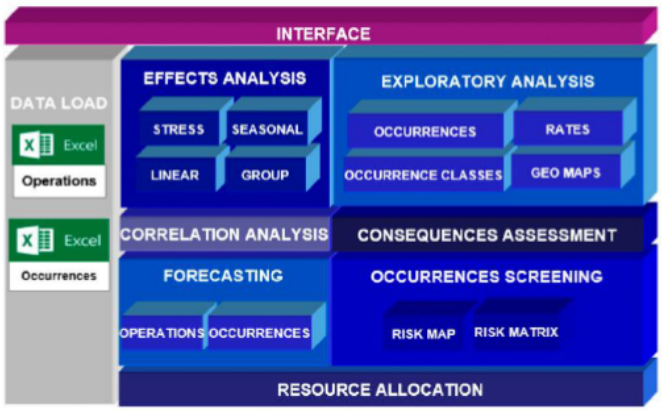
Bayesian methods and official stats. RIMAS @AESA

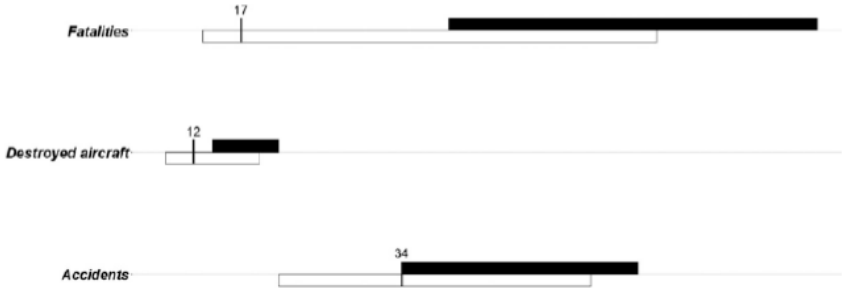


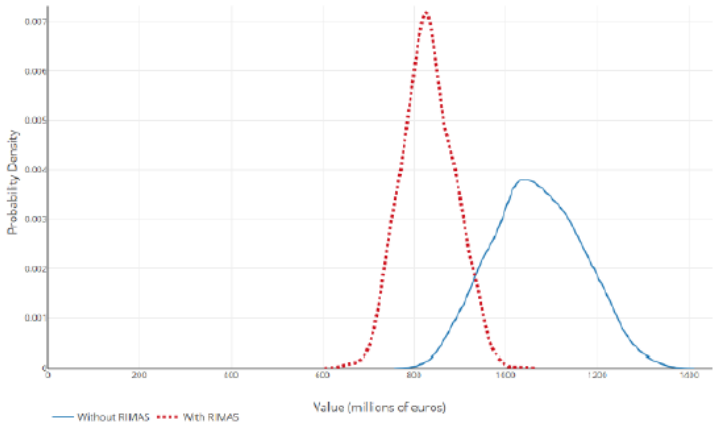


2000+ forecasting models

Bayesian methods and official stats. RIMAS @AESA



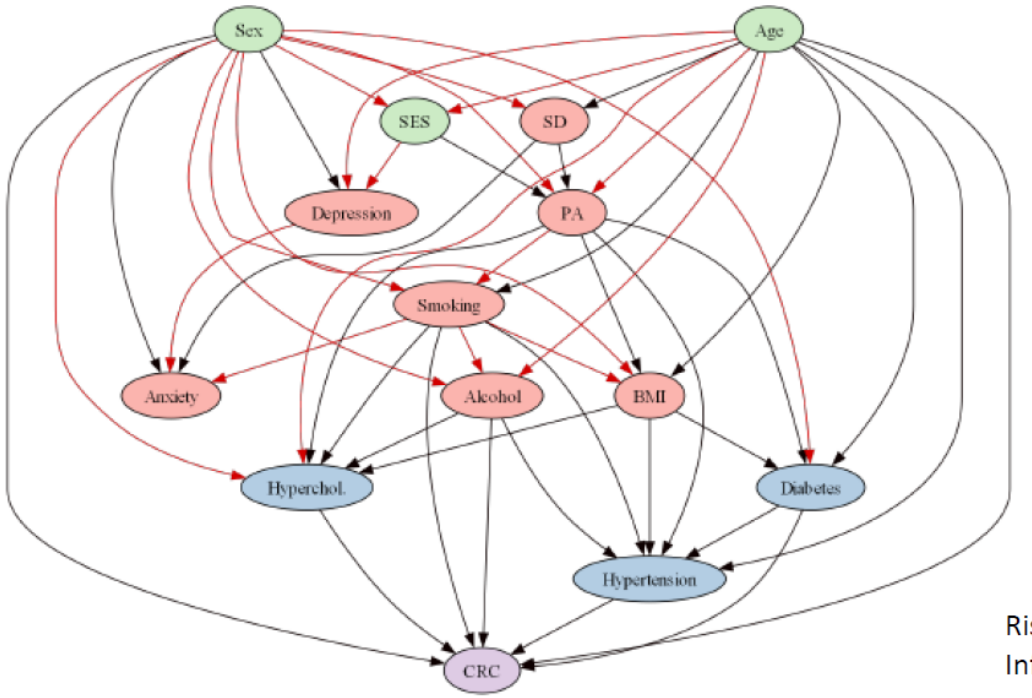




800 M€ safety equivalent annual savings

Bayesian methods and official stats: ONCOSCREEN HE

$$
\begin{aligned}
p(v_{sex},\ldots,v_{depression}) = {} & \Big[p(v_{sex})p(v_{age})p(v_{SES}|v_{sex},v_{age})\Big]\times \\
& \Big[p(v_{SD}|v_{sex},v_{age})p(v_{PA}|v_{sex},v_{age},v_{SD},v_{SES})p(v_{depr}|v_{sex},v_{age},v_{SES}) \\
& p(v_{smok}|v_{sex},v_{age},v_{PA})p(v_{alc}|v_{sex},v_{age},v_{smok}) \\
& p(v_{BMI}|v_{sex},v_{age},v_{PA},v_{smok})p(v_{anx}|v_{sex},v_{SD},v_{smok},v_{depr})\Big]\times \\
& \Big[p(v_{hypchol}|v_{sex},v_{age},v_{PA},v_{smok},v_{BMI},v_{alc})p(v_{diab}|v_{sex},v_{age},v_{PA},v_{BMI}) \\
& p(v_{hypten}|v_{age},v_{PA},v_{smok},v_{BMI},v_{alc},v_{diab})\Big]\times \\
& p(v_{CRC}|v_{sex},v_{age},v_{alc},v_{smok},v_{hypchol},v_{hypten},v_{diab}) \qquad (1)
\end{aligned}
$$



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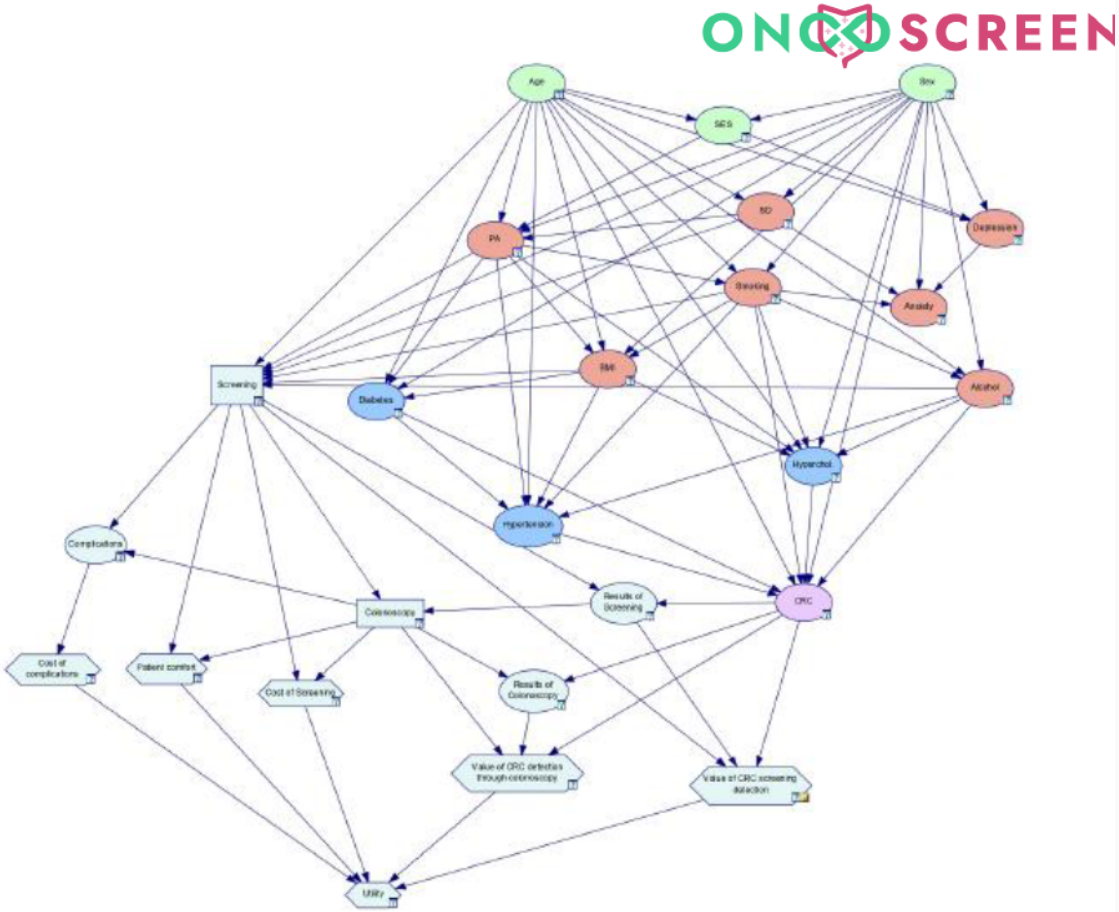
Risk maps
Influencial variables

Bayesian methods and official stats: ONCOSCREEN HE

Cost intervention
Cost complications
Comfort
Information

MC Value function
Risk averse utility function



Bayesian methods and official stats: ONCOSCREEN HE

- Individual recommendations. Personalised screening based on risk
- Benchmarking new screening devices (ONCOSCREEN)
- Assessing screening programs (and designing)

Current Spanish Strategy	Proposed Strategy
• If patient >50 years old: Send FIT invitation regardless their features. • Else: No screening.	For patient with features X: • if $p(CRC\|X) < \theta_1 \rightarrow$ No screening • if $\theta_1 \leq p(CRC\|X) < \theta_2 \rightarrow$ FIT • if $p(CRC\|X) \geq \theta_2 \rightarrow$ sDNA

Benefits of Bayesian Stats

Benefits of Bayesian Stats

- Simpler interpretations
- More natural concepts
- Probability distributions for observations and parameters
- Integration of expert opinion
- Apportioning of all sources of uncertaimty
- Integration with decision making

Course logistics

Course logistics

- Schedule. Mon13-Fri17. 10:00-13:00 through Webex
- Materials and labs @ `https://icmat-datalab.github.io/intro_estadistica_bayesiana_ine/`
- Discord `https://discord.gg/fk3UQFge`
- David david.rios.insua@gmail.com
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- Mario mario.chacon@icmat.es

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Lab1

Lab1

A brief intro/reminder of R